Typing Judgments for Stack-Based Concatenative Languages

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1 Subsumption

Denote by $\prec$ the subsumption relation. A sum subsumes its constituents, i.e.

$$A \prec A \oplus B$$
$$B \prec A \oplus B$$

Subsumption is contravariant in the argument to a function type: $A - -B \prec A' - -B'$ iff $A' \prec A$ and $B' \prec B$; a function is more general if it accepts a wider type.

2 Judgments

Lowercase Greek letters are type variables; uppercase Latin letters are stack variables. Γ is a context. Lowercase Latin letters are terms.

A stack variable A is a sequence of type variables $\alpha_1, \alpha_2, \ldots, \alpha_n$ of undetermined length.

$$\frac{\Gamma \vdash x : A - -B\ C \qquad \Gamma \vdash y : C' - -D \qquad C \prec C'}{\Gamma \vdash xy : A - -B\ D} \quad (\textsc{Cut})$$

$$\frac{\Gamma \vdash x : A - -B}{\Gamma \vdash [x] : C - -C\ [A - -B]} \quad (\textsc{Quote})$$

$$\frac{}{\Gamma \vdash \texttt{apply} : A\ [A - -B] - -B} \quad (\textsc{Apply})$$

Note that stack variables are always leftmost, hence the inconvenient (from the perspective of the programmer) ordering of `apply`.

Tags in Pizarnik have an associated arity.

$$\Gamma \vdash x_0 : A`\beta_1\beta_2 \cdots \beta_n\alpha_0 - -C_0 \qquad x_1 : AB_1 - -C_1 \quad (\textsc{Pattern-Match})$$

3 Unification

The above is declarative, rather than algorithmic typing.

The Cut rule above only requires subsumption rather than equality. Hence we introduce $\alpha \uplus_{\lhd} \beta \Rightarrow s$, "$\alpha$ is subsumed by β under substitution s."

3.1 Monomorphization